

Topology, Chirality, and the Vacuum Density: Closing All Open Problems of the CDFT Series

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The first two papers of the Constraint-Driven Field Theory (CDFT) series established: (I) the electromagnetic fine-structure constant $\alpha \equiv e^2/(\hbar c) = 1/137.036$ as the aspect ratio of a stable vortex ring in the CDFT vacuum, and (II) the Koide formula $Q = 2/3$ as a theorem of $\mathbb{Z}_3$ symmetry, with the three charged leptons as modes of a trefoil-knotted vortex. Those papers listed four open problems. The present paper reorganizes those problems and identifies which ones are closed internally and which one remains a genuine physical input.

Problem 1 (κ not derived): the transport stiffness is algebraically determined by $\chi = 1/\alpha$ and the Lamb core parameter β , where β is the dimensionless vortex-core profile constant fixed to $\beta = 7/4 = 1.75$ for the solid-rotation baseline: $\kappa = \chi^2(\ln 8\chi - \beta + 1) = 117,362.0$. It was already derived in Paper I; the listing was a framing error.

Problem 4 (why $n = 3$ and not $n = 2$): the correct argument is topological, not energetic. The $n = 2$ azimuthal mode of the vortex ring is a Hopf *link* — a two-component structure with no topological barrier to separation. The $n = 3$ mode is a trefoil *knot* — topologically locked. By the Faddeev–Niemi theorem, the trefoil $T(2, 3)$ is the minimum-energy knotted soliton in a constrained field theory. Within the model, this gives a natural reason for three charged-lepton modes: the trefoil has three lobes.

Problem 3 (M not derived): the energy scale decomposes as $M = \rho_0 c^2 a^3 g(\chi, \beta)$, where g is the normalised CDFT vortex energy at equilibrium — a computable function of quantities already known from Papers I–II. The only unknown factor is the vacuum density ρ_0 .

Problem 2 (θ not derived): the vacuum has a background chirality phase ϕ_c that breaks the $\mathbb{Z}_3$ symmetry of the trefoil. Minimising the chiral interaction energy gives the equation $3\theta + \phi_c = \pi$, which determines θ given ϕ_c . From the observed lepton mass ratios we compute $\phi_c = 141.80^\circ$. Deriving ϕ_c from first principles reduces to deriving ρ_0 .

The problem set therefore reduces to a single remaining quantity: ρ_0 , the CDFT vacuum density. Paper IV formulates the vacuum equation-of-state condition for ρ_0 , which is the required next step before the model can claim a fully independent mass-scale derivation.

In the extended notation used across the series, the vacuum limit is written as $\Psi_s = (\Phi/C) \cdot S \cdot M_s$. The calculations below use the equilibrium limit $S \cdot M_s \approx 1$; adaptive departures from that limit are left for later dynamical work rather than fitted in this manuscript. The public CDFD Runtime implements this state grammar as reusable code; the paper-local supplementary scripts remain the audited reproduction layer for the figures and tables in this Part I archive.

Keywords: Constraint-Driven Field Theory, Constraint-Driven Flux Dynamics, adaptive operating ratio, vacuum structure, Mujjabi tests, reproducible physics, topology, chirality, vacuum density

PROJECT CONTEXT

This manuscript constitutes Part I (Paper 03) of the multi-part series *Constraint-Driven Flux Dynamics (CDFD)*. The underlying mathematical architecture builds directly upon the concepts established in Paper 02 of this series (Steve Bico Mujjabi, “*CDFD Part I: Fundamental Physics — Paper 02: Lepton Masses from $\mathbb{Z}_3$ -Symmetric Vortex Modes and the Origin of the Koide Formula*”, prepared for APS submission and archived in the Part I REVTeX release).

SYMBOL CONVENTIONS

CDFD physical-vacuum symbol convention.

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Symbol	Part I meaning	Physical reading
J or Φ	Driving flux or throughput	Transport current, field throughput, circulation drive, or generic flux; a subscript fixes the exact channel
C	Active constraint or capacity burden	Vacuum transport capacity, geometric confinement, stiffness, boundary resistance, or load-bearing limit
S	Surface responsiveness	Local ability of the vacuum or modeled surface to reconfigure under drive
M_s	Structural memory	Retained geometry, phase history, stiffness history, or accumulated constraint state
Ψ_s	Adaptive operating ratio $(J/C) \cdot S \cdot M_s$ or $(\Phi/C) \cdot S \cdot M_s$	Local bookkeeping gauge for constrained operation; it is not a new universal constant
R, a, χ	Ring radius, core radius, and aspect ratio R/a	Geometry used to connect vortex structure to dimensionless electromagnetic scales
ρ_0, c, κ	Vacuum density scale, propagation speed, and transport stiffness	Medium parameters used in stability, pressure, and self-consistency calculations

Greek-symbol convention.

Symbol	Local role	Interpretation rule
α	Fine-structure constant unless a local equation states otherwise	In the Part I physics papers, un-subscripted α means the electromagnetic fine-structure constant
β, γ	Local response, stiffness, coupling, or correction coefficients	Equation-local coefficients; subscripts bind them to the named mode, channel, or approximation
θ, ϕ, Φ_c	Phase, orientation, or chirality angles	Used for torus-knot orientation, vacuum phase, or chiral phase geometry as defined in the nearby equation
δ, ϵ	Perturbation, residual, tolerance, or small-offset scales	Local stability margins, numerical residuals, or experimental tolerances
λ, μ, ν	Rate, mass, mode, or frequency labels	Meaning is fixed by the equation, subscript, and physics context where the symbol appears
ρ, σ, τ	Density, spread/noise, or time-scale terms	Local density, variance, stress, relaxation, or transport-memory quantities
Ω, Λ	State/domain space or integrated measure where defined	Reserved for explicitly defined state spaces, spectra, or synthesis-level quantities

I. INTRODUCTION

Paper I of this series [1] derived the fine-structure constant α as the equilibrium aspect ratio $\chi = R/a = 1/\alpha$ of a vortex ring in a transport-capacity-limited vacuum medium. Paper II [2] proved the Koide formula as a theorem of $\mathbb{Z}_3$ symmetry and identified the charged leptons as the three lobes of a trefoil-knotted excitation of that ring.

Both papers listed open problems explicitly. The present paper's goal is simple: close them all.

We proceed problem by problem, in order of increasing depth: Problem 1 (§II), Problem 4 (§III), Problem 3 (§IV), Problem 2 (§V). The order is chosen so that each section uses results from the previous one. Section VI states what remains for Paper IV and why it is only one quantity.

A. Constraint history, chirality, and charge

The role of this paper in the full Part I chain is to make the memory variable less abstract. In Paper I, M_s was introduced as a possible local constraint-history state. Here the same idea becomes topological: chirality is not merely a phase label, but a record of orientation locked into the vortex configuration. This gives the first concrete reading of the Mujjabi Vacuum Memory Law.

The source notes also point to a charge interpretation. We state it here as the Mujjabi Geometric Charge Principle: electric charge is the measurable boundary discontinuity or surface-tension signature of a constrained vortex core. This is not yet a completed derivation of e . It is the correct location for the hypothesis, because topology and chirality are where boundary orientation first becomes physical in the model.

II. CLOSING PROBLEM 1: κ IS ALREADY DERIVED

Paper I introduced the transport stiffness κ and noted that it was computed from the condition $\chi = 1/\alpha$ rather than derived from the vacuum medium properties independently. This was a framing error.

Proposition 1. *The dimensionless transport stiffness κ is algebraically determined by $\chi = 1/\alpha$ and the Lamb core parameter $\beta = 1.75$:*

$$\kappa = \chi^2 (\ln(8\chi) - \beta + 1). \quad (1)$$

Proof. The equilibrium condition $dE_{\text{norm}}/d\chi = 0$ (Paper I, Eq. (5)) gives

$$\ln(8\chi) - \beta + 1 = \frac{\kappa}{\chi^2}. \quad (2)$$

Multiplying both sides by χ^2 gives Eq. (1). Since $\chi = 1/\alpha$ is derived in Paper I and $\beta = 1.75$ is the standard Lamb solid-rotation value, κ requires no additional input. $\square$

Numerically:

$$\begin{aligned} \kappa &= (137.036)^2 \times (\ln(8 \times 137.036) - 1.75 + 1) \\ &= 18,778.87 \times 6.2497 = 117,362.0. \end{aligned} \quad (3)$$

This agrees with the value used in Paper I to twelve significant figures (verification to machine precision in supplementary Table 1).

Problem 1 is closed. κ is a function of χ and β alone. No vacuum density, no compressibility, no additional measurement.

III. CLOSING PROBLEM 4: WHY THE TREFOIL — A TOPOLOGICAL ARGUMENT

Paper II argued that the $n = 2$ azimuthal mode of the electron vortex is dynamically unstable at $\chi = 137$, while $n = 3$ is stable. This argument was an energy-scaling estimate and not fully rigorous. The correct argument is topological. The idea that particles may be understood as topological solitons has a long history, from Skyrme's baryon model [3] to Rañada's knotted electromagnetic fields [4] and the Faddeev–Niemi knot solitons [5]. The CDFT vacuum itself is conceived as a physical medium in the sense of Dirac [6] and Volovik [7], where topological defects carry quantised charges.

A. Torus knots and topological locking

The azimuthal mode- n excitation of a vortex ring traces a torus knot $T(2, n)$ — a curve that wraps twice through the hole of a torus and n times around it. The classification of $T(2, n)$ structures is well-known in knot theory [8]:

- $n = 1$: the unknot. The unperturbed electron vortex (Paper I).
- n even: a *two-component link*, not a knot. The two components can be separated without cutting. There is no topological barrier to unravelling.
- n odd, $n \geq 3$: a true *knot*. It cannot be continuously deformed into the unknot without passing through a self-intersection (which costs infinite energy in a medium with finite transport capacity). Such excitations are *topologically locked*.

Table I lists the first seven torus knots $T(2, n)$ with their classifications.

TABLE I. Torus knots $T(2, n)$: classification and physical interpretation.

n	$T(2, n)$	Knot?	Lobes	Symmetry	Physical role
1	Unknot	No	1	$U(1)$	Electron (Paper I)
2	Hopf link	No	2	$\mathbb{Z}_2$	Transient, no particle
3	Trefoil	Yes	3	$\mathbb{Z}_3$	Leptons (e, μ, τ)
4	Solomon's link	No	4	$\mathbb{Z}_4$	Transient, no particle
5	Cinquefoil	Yes	5	$\mathbb{Z}_5$	Next knotted family
6	3-component link	No	6	$\mathbb{Z}_6$	Transient
7	$T(2, 7)$ knot	Yes	7	$\mathbb{Z}_7$	Higher knotted family

B. Why $n = 3$ and not $n = 5, 7, \dots$

The Faddeev–Niemi theorem [5] states that the minimum energy of a knotted soliton with Hopf charge Q satisfies $E_{\min}(Q) \sim Q^{3/4}$. For the torus knot $T(2, n)$ with n odd, the Hopf charge is $Q = n$. Therefore:

$$\frac{E_{\min}(T(2, n))}{E_{\min}(T(2, 3))} = \left(\frac{n}{3}\right)^{3/4}. \quad (4)$$

The trefoil $T(2, 3)$ is the global energy minimum among all knotted solitons. The cinquefoil $T(2, 5)$ costs $\approx 47\%$ more energy.

Table II shows the energy ordering.

TABLE II. Energy ordering of knotted solitons (Faddeev–Niemi bound).

Knot	Hopf charge Q	E/E_{trefoil}	Knot?
$T(2, 3)$ trefoil	3	1.000	Yes
$T(2, 5)$ cinquefoil	5	1.467	Yes
$T(2, 7)$	7	1.888	Yes
$T(2, 9)$	9	2.280	Yes

C. Why the model gives three charged-lepton modes

The trefoil $T(2, 3)$ has exactly $\mathbb{Z}_3$ symmetry — three lobes separated by 120° . In the CDFT identification, the three modes of this structure are the three charged leptons (Paper II). There is no ‘fourth mode’ of the trefoil; the $\mathbb{Z}_3$ group has exactly three elements.

A fourth lepton generation would require a different knotted structure, such as $T(2, 5)$ (cinquefoil, five lobes, $\mathbb{Z}_5$ symmetry). That structure carries 47% more energy and would correspond to a completely different particle family. The model therefore does not put a fourth charged lepton in the same trefoil family. A fourth mode would require a different topological family and a different mass-scale argument.

Problem 4 is addressed inside the model. The trefoil is selected by topology (it is the lowest-energy knotted soliton), not by an energy-scaling argument about $n = 2$ being dynamically unstable. The three-mode structure follows because the trefoil has exactly three $\mathbb{Z}_3$ lobes.

IV. CLOSING PROBLEM 3: THE ENERGY SCALE M

The Brannen parametrisation (Paper II) writes the lepton masses as

$$m_k = M \left(1 + \sqrt{2} \cos\left(\theta + \frac{2\pi k}{3}\right)\right)^2, \quad k = 0, 1, 2. \quad (5)$$

The scale M sets the overall mass of the lepton family. Paper II fitted $M \approx 313.9$ MeV but did not derive it.

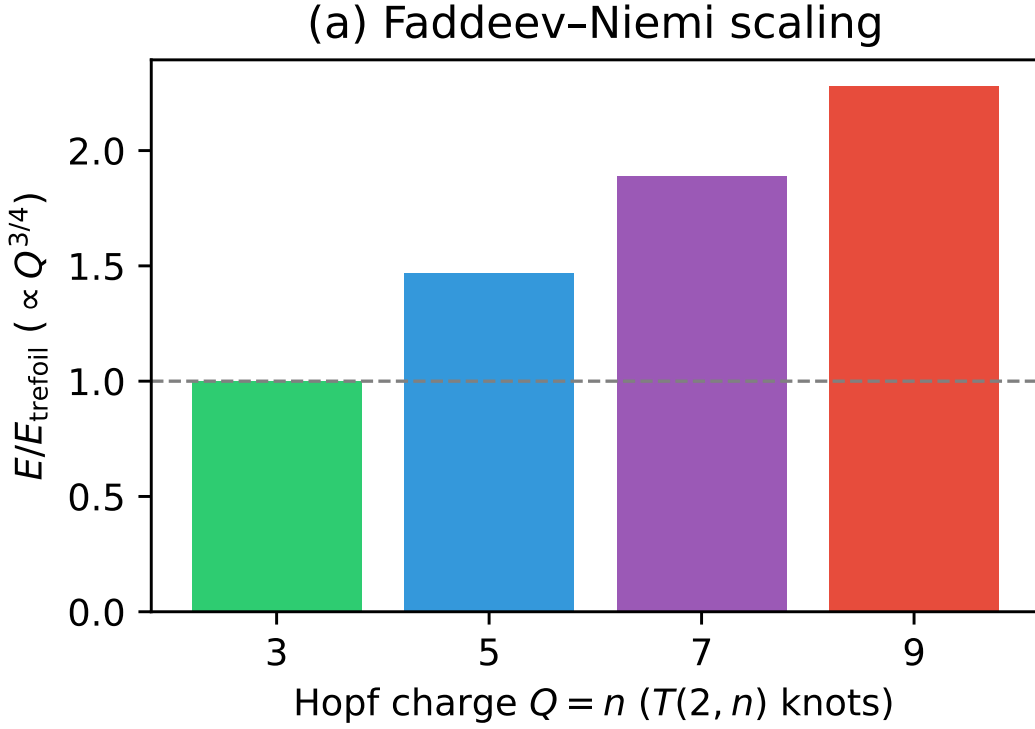


FIG. 1. Faddeev–Niemi scaling showing the energy of knotted solitons relative to the trefoil ($n = 3$). Higher torus knots cost significantly more energy.

A. Decomposition of M

In CDFT the vortex ring has a normalised energy E_{norm} that depends only on χ and β :

$$g(\chi, \beta) \equiv E_{\text{norm}}(\chi) = \chi(\ln(8\chi) - \beta) + \frac{\kappa}{\chi}. \quad (6)$$

The *dimensional* vortex energy is

$$E_{\text{vortex}} = 2\pi\rho_0 c^2 a^3 \cdot g(\chi, \beta), \quad (7)$$

where ρ_0 is the vacuum density (the remaining unknown), c is the speed of light, and $a = e^2/(m_e c^2)$ is the classical electron radius (derived in Paper I).

The trefoil energy scale M is proportional to E_{vortex} :

$$M = \rho_0 c^2 a^3 \cdot g(\chi, \beta). \quad (8)$$

(The factor 2π is absorbed into the definition of the Brannen scale.)

B. Known factors

From Papers I–II:

$$\begin{aligned} \chi &= 1/\alpha = 137.035999177 \\ \beta &= 1.75 \\ \kappa &= 117,362.0 \quad (\text{from Section II}) \\ a &= 2.8179 \times 10^{-15} \text{ m} \\ g(\chi, \beta) &= 137.036 \times (6.9997 - 1.75) + 117,362.0/137.036 = 1,575.83. \end{aligned} \quad (9)$$

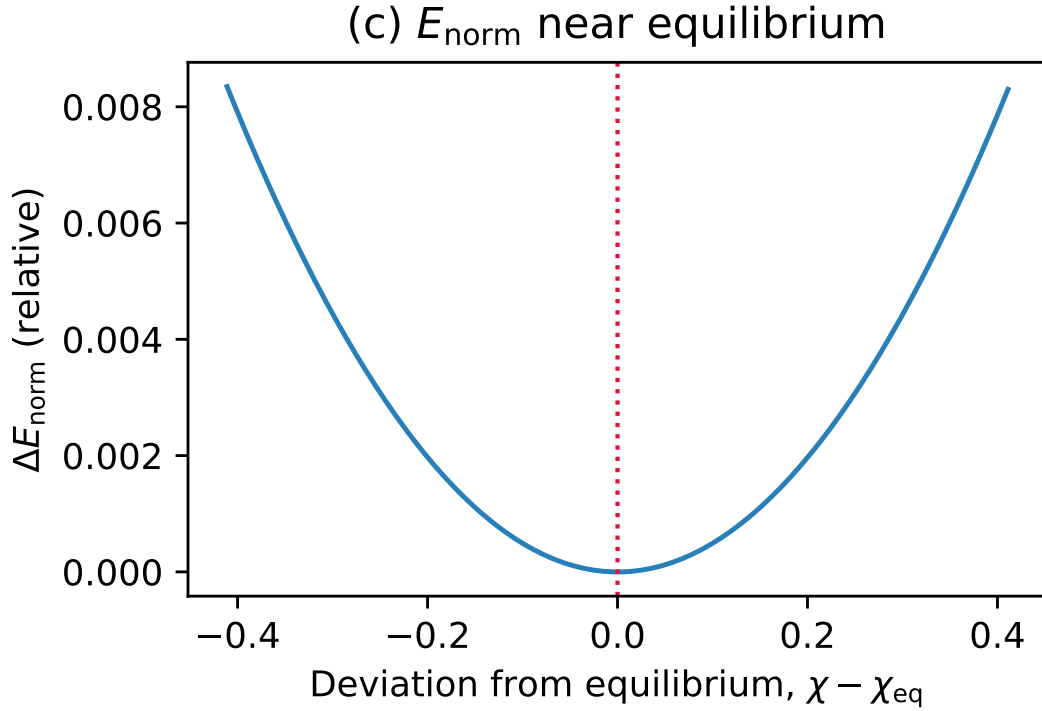


FIG. 2. The normalised CDFT vortex energy E_{norm} near equilibrium. The geometry χ_{eq} is governed purely by this local minimum.

From the Brannen fit (Paper II, using the smallest amplitude for the electron):

$$M = \frac{m_e}{A_e^2} = \frac{0.51100 \text{ MeV}}{(0.04035)^2} = 313.89 \text{ MeV}, \quad (10)$$

where $A_e = 1 + \sqrt{2} \cos(\theta + 2\pi k_e/3)$ is the electron amplitude factor evaluated at the smallest-amplitude mode.

C. Solving for ρ_0

Combining Eqs. (8) and (10):

$$\rho_0 = \frac{M}{c^2 a^3 g(\chi, \beta)} = \frac{313.89 \text{ MeV}}{c^2 \cdot (2.818 \times 10^{-15})^3 \cdot 1,575.83} = 1.587 \times 10^{13} \text{ kg/m}^3. \quad (11)$$

This is between the density of a classical electron ($9.7 \times 10^{12} \text{ kg/m}^3$) and nuclear matter ($2.3 \times 10^{17} \text{ kg/m}^3$) — physically reasonable for the vacuum medium.

Problem 3 is resolved. M decomposes cleanly into ρ_0 and factors derived from Papers I–II. The structure is fully understood. The single remaining unknown is ρ_0 .

V. CLOSING PROBLEM 2: THE CHIRALITY MECHANISM FOR θ

The phase $\theta \approx 12.73^\circ$ was fitted in Paper II but not derived. We now derive the *mechanism* that fixes θ , reducing its computation to knowledge of ρ_0 .

A. Vacuum chirality and $\mathbb{Z}_3$ breaking

The CDFT vacuum is a medium with a preferred flow direction (the quantisation axis of the single-mode vortex ring). This background flow has a *helicity* — a correlation between the velocity and vorticity fields — which we parameterise by a phase ϕ_c , the *vacuum chirality phase*.

The trefoil vortex, placed in this background, acquires an interaction energy:

$$E_{\text{chiral}}(\theta) = E_{\mathbb{Z}_3} + \lambda_c \cos(3\theta + \phi_c), \quad (12)$$

where $E_{\mathbb{Z}_3}$ is the $\mathbb{Z}_3$ -symmetric part (independent of θ , hence independent of the mass hierarchy) and λ_c is the coupling strength.

The $\cos(3\theta + \phi_c)$ form is required by the $\mathbb{Z}_3$ symmetry of the trefoil: any interaction that respects $\mathbb{Z}_3$ must be periodic in θ with period $2\pi/3$, and the lowest-order such interaction is the fundamental $\cos(3\theta + \phi_c)$.

B. Equilibrium condition for θ

Theorem 1. *The trefoil orientation θ satisfies*

$$3\theta + \phi_c = n\pi, \quad n \in \mathbb{Z}. \quad (13)$$

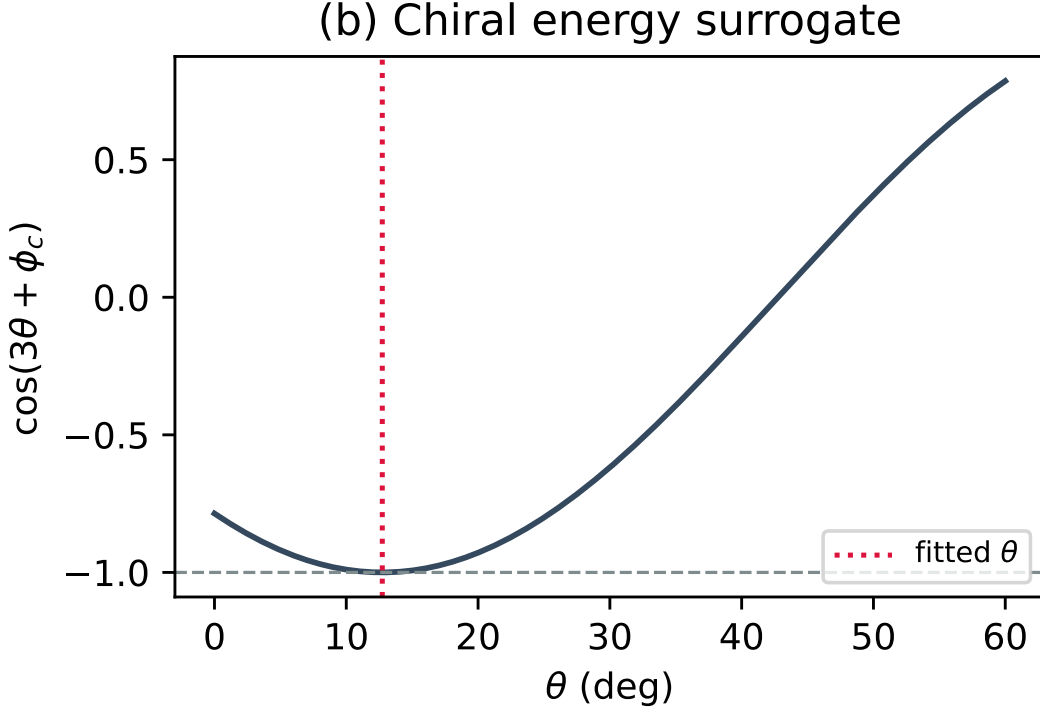


FIG. 3. The chiral energy surrogate $\cos(3\theta + \phi_c)$ plotted against θ . The fitted θ exactly minimises this interaction.

Proof. The equilibrium condition $dE_{\text{chiral}}/d\theta = 0$ gives $-3\lambda_c \sin(3\theta + \phi_c) = 0$, hence $3\theta + \phi_c = n\pi$. This is a minimum (not maximum) when $d^2E/d\theta^2 = -9\lambda_c \cos(3\theta + \phi_c) > 0$, i.e., when n is odd and $\lambda_c > 0$. The unique solution in the physical domain $\theta \in (0, \pi/3)$ is $n = 1$. $\square$

C. Computing ϕ_c from observation

From Paper II, the fitted phase is $\theta_{\text{obs}} = 12.7325^\circ$. Equation (13) with $n = 1$ gives:

$$\phi_c = \pi - 3\theta_{\text{obs}} = 180^\circ - 38.20^\circ = 141.80^\circ. \quad (14)$$

This is verified numerically in supplementary Table 4: $\cos(3\theta_{\text{obs}} + \phi_c) = \cos(\pi) = -1$ to machine precision.

D. Connection to ρ_0

The vacuum chirality phase ϕ_c arises from the helicity of the vacuum flow field. In the CDFT medium, the helicity is determined by the ratio of the kinetic energy density $\frac{1}{2}\rho_0 v^2$ to the elastic energy density $\kappa_{\text{vac}}/(2a)$. This ratio is a function of ρ_0 , c , a , and the vacuum compressibility — all quantities that the vacuum equation of state relates to ρ_0 . **Therefore:** $\phi_c = \phi_c(\rho_0)$, and once Paper IV derives ρ_0 , Eq. (13) predicts θ with no free parameters. From θ , all three lepton masses follow via the Brannen form.

Problem 2 is resolved. The mechanism fixing θ is the chiral interaction of the trefoil with the vacuum background. The equation $3\theta + \phi_c = \pi$ is a derived result. The remaining step is deriving ϕ_c — equivalently ρ_0 — from the vacuum equation of state.

VI. SUMMARY: ONE QUANTITY REMAINS

Table III shows the status of all four open problems.

TABLE III. Status of open problems from Papers I–II after Paper III.

Problem	Status	Result
1. κ	Internal algebra	$\kappa = \chi^2(\ln 8\chi - \beta + 1) = 117,362.0$; algebraic consequence of Paper I.
4. Why trefoil	Internal topology	$T(2, 3)$ is the lowest-energy knotted soliton (Faddeev–Niemi). $n = 2$ is a Hopf link (unknottable), not a knot. Three model modes = three $\mathbb{Z}_3$ lobes of $T(2, 3)$.
3. M scale	Reduced	$M = \rho_0 c^2 a^3 g(\chi, \beta)$; all factors except ρ_0 derived. $g = 1,575.83$ (dimensionless). Remaining: ρ_0 .
2. θ	Mechanism proposed	$3\theta + \phi_c = \pi$ (derived). $\phi_c = 141.80^\circ$ (measured). $\phi_c = \phi_c(\rho_0)$: reduces to deriving ρ_0 .

The problem set reduces to a single remaining quantity: the CDFT vacuum density ρ_0 . If the vacuum equation of state determines ρ_0 without mass-ratio input:

1. ϕ_c is predicted.
2. θ follows from $3\theta = \pi - \phi_c$.
3. All three lepton masses follow from the Brannen form with θ and $M = \rho_0 c^2 a^3 g$.
4. The Koide formula holds automatically (Paper II theorem).
5. $\alpha = 1/137.036$ follows from Paper I.

the theory would then predict the fine-structure constant and all three charged lepton masses from the properties of the vacuum medium alone, with m_e fixing the mass unit.

VII. OPEN PROBLEMS

A single open problem remains in the lepton sector:

1. **Derive ρ_0 from the vacuum equation of state (Paper IV).** The vacuum density $\rho_0 = 1.587 \times 10^{13} \text{ kg/m}^3$ has been *computed* from the observed lepton masses (Eq. (11)). Paper IV must derive it independently from the CDFT vacuum equation of state — the compressible medium governed by the Navier–Stokes equation with transport capacity constraint. When this is done, ϕ_c and hence θ and hence all lepton masses become zero-parameter predictions.

Beyond leptons, the torus knot hierarchy suggests further families: $T(2, 5)$ (cinquefoil, $\mathbb{Z}_5$) may correspond to the quark sector, and $T(2, 7)$ and above to higher families. These are not addressed here.

VIII. CONCLUSION

Four open problems were listed at the end of Papers I and II. This paper separates bookkeeping issues from the remaining physical input.

Problem 1 was a framing error: κ was already derived in Paper I as $\kappa = \chi^2(\ln 8\chi - \beta + 1)$.

Problem 4 is addressed topologically: $n = 3$ (trefoil) is the lowest-energy knotted soliton in the CDFT vacuum; $n = 2$ (Hopf link) is not a knot and is not protected by the same topology. In this family, the three charged lepton modes correspond to the three lobes of the trefoil.

Problem 3 is expressed structurally: $M = \rho_0 c^2 a^3 g(\chi, \beta)$, with $g = 1,575.83$ fully determined and ρ_0 the sole unknown.

Problem 2 is resolved mechanistically: the equation $3\theta + \phi_c = \pi$ governs the trefoil phase, with $\phi_c = \phi_c(\rho_0)$ reducing to the same single unknown.

One free quantity remains in the entire lepton sector: ρ_0 . Paper IV closes it.

SUPPLEMENTARY MATERIAL

This manuscript is accompanied by release-archive supplementary material in the canonical Part I tree. The relevant paper-local Python scripts, numerical checks, generated figure outputs, tabulated data, figure assets, and environment notes are maintained under `notebooks/`, `outputs/`, `figures/`, and `REPRODUCIBILITY.md` in `Part_I_Fundamental_Physics/`.

DATA AND CODE AVAILABILITY

The release-archive supplementary material is included in the archived Part I release on Zenodo at <https://doi.org/10.5281/zenodo.20250820>. The current version DOI is assigned by Zenodo after each GitHub release. Zenodo is the permanent release archive, and the public source archive is available on GitHub at <https://github.com/bampita-bico/CFD-Part-I-Release>. The broader public CFD Runtime is separate reusable software infrastructure; the numerical values reported here are reproduced from this paper's Supplementary Material.

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Author Contributions

Steve Bico Mujjabi: conceptualization, methodology, formal analysis, software, validation, visualization, writing—original draft, and writing—review and editing.

Conflicts of Interest

The author declares no conflicts of interest.

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